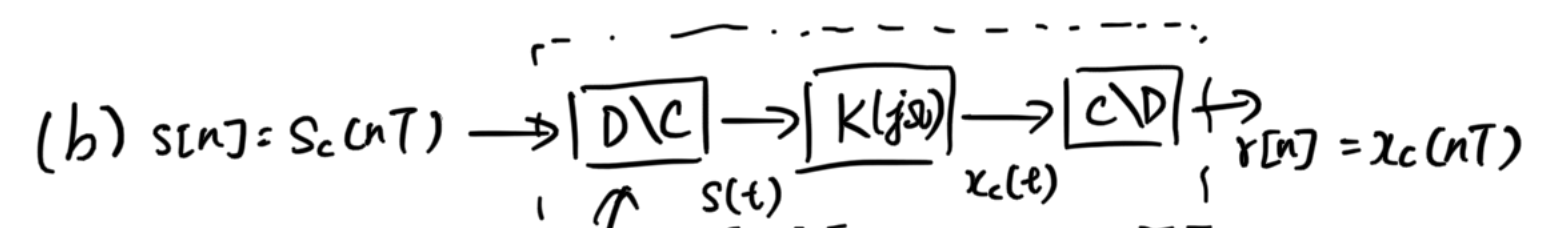


4.7 (a) $X_c(j\Omega) = S_c(j\Omega) + \alpha S_c(j\Omega) e^{-j\Omega\tau_d}$

$= (1 + \alpha e^{-j\Omega\tau_d}) S_c(j\Omega)$

$X(e^{j\omega}) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - \frac{2\pi n}{T}))$
 $= \frac{1}{T} \sum_{n=-\infty}^{\infty} (1 + \alpha e^{-j(\frac{\omega}{T} - \frac{2\pi n}{T})\tau_d}) S_c(j(\frac{\omega}{T} - \frac{2\pi n}{T}))$



$S(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} S_c(j(\frac{\omega}{T} - \frac{2\pi k}{T}))$

$H(e^{j\omega}) = 1 + \alpha e^{-j(\frac{\omega}{T} - \frac{2\pi k}{T})\tau_d}$

$= 1 + \frac{\alpha}{2\pi} \sum_{k=-\infty}^{\infty} \int_{-\pi}^{\pi} e^{-j\omega\frac{\tau_d}{T}} \cdot e^{-j\frac{2\pi k}{T}\tau_d} d\omega$

$= 1 + \alpha \cdot \sum_{k=-\infty}^{\infty} \frac{\sin[(\frac{\tau_d}{T} + k)\pi]}{(\frac{\tau_d}{T} + k)\pi} e^{j\frac{2\pi k}{T}\tau_d}$

$= 1 + \alpha \sum_{k=-\infty}^{\infty} S_a(\frac{\tau_d}{T} + k) e^{j\frac{2\pi k}{T}\tau_d}$

(c) $h[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} (1 + \alpha e^{-j\omega\frac{\tau_d}{T}}) e^{j\omega n} d\omega$

$= \frac{1}{2\pi} \cdot \alpha \int_{-\pi}^{\pi} e^{j\omega(n - \frac{\tau_d}{T})} d\omega + \delta[n]$

(i) $\tau_d = T$

$h[n] = \delta[n] + \alpha \delta[n-1]$

(ii) $\tau_d = \frac{T}{2}$

$h[n] = \frac{\alpha}{2\pi} \cdot \frac{1}{j(n - \frac{1}{2})} (e^{j\pi(n - \frac{1}{2})} - e^{-j\pi(n - \frac{1}{2})}) + \delta[n]$

$= \frac{\alpha}{2\pi} \cdot \frac{2j \sin(\pi(n - \frac{1}{2}))}{j(n - \frac{1}{2})} + \delta[n]$

$= \alpha \cdot \frac{\sin(\pi(n - \frac{1}{2}))}{\pi(n - \frac{1}{2})} = \alpha S_a(n - \frac{1}{2}) + \delta[n]$

4.15 $X_r(e^{j\omega}) = H(e^{j\omega}) X_e(e^{j\omega}) = H(e^{j\omega}) X_d(e^{j\omega}) = \frac{1}{M} H(e^{j\omega}) \sum_{k=0}^{M-1} X(e^{j(\omega - \frac{2\pi k}{M})})$

$= \begin{cases} \sum_{k=0}^{M-1} X(e^{j(\omega - \frac{2\pi k}{M})}) & |\omega| \leq \frac{\pi}{3} \\ 0 & |\omega| \in (\frac{\pi}{3}, \pi] \end{cases}$

(a) $x[n] = \cos(\pi n/4) = \frac{1}{2} (e^{j\frac{\pi}{4}n} + e^{j(-\frac{\pi}{4})n})$

$X(e^{j\omega}) = \frac{1}{2} \cdot 2\pi \left(\sum_{k=-\infty}^{\infty} \delta(\omega - \frac{\pi}{4} - 2\pi k) + \sum_{k=-\infty}^{\infty} \delta(\omega + \frac{\pi}{4} - 2\pi k) \right)$

$\omega_N = \frac{\pi}{4} \leq \frac{\pi}{3} \Rightarrow x_r[n] = x[n]$

(b) $x[n] = \cos(\pi n/2)$

$\Rightarrow \omega_N = \frac{\pi}{2} > \frac{\pi}{3} \Rightarrow x_r[n] \neq x[n]$

(c) $g[n] = \frac{\sin(\pi n/8)}{\pi n/8} \Leftrightarrow G(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{8} \\ 0, & \text{otherwise} \end{cases}$

$x[n] = 64 g[n]^2 \Leftrightarrow X(e^{j\omega}) = 64 \cdot \frac{1}{2\pi} G(e^{j\omega}) * G(e^{j\omega}), (|\omega| \leq \pi)$

$= \frac{64}{2\pi} \int_{-\pi}^{\pi} G(e^{j\theta}) G(e^{j(\omega - \theta)}) d\theta$

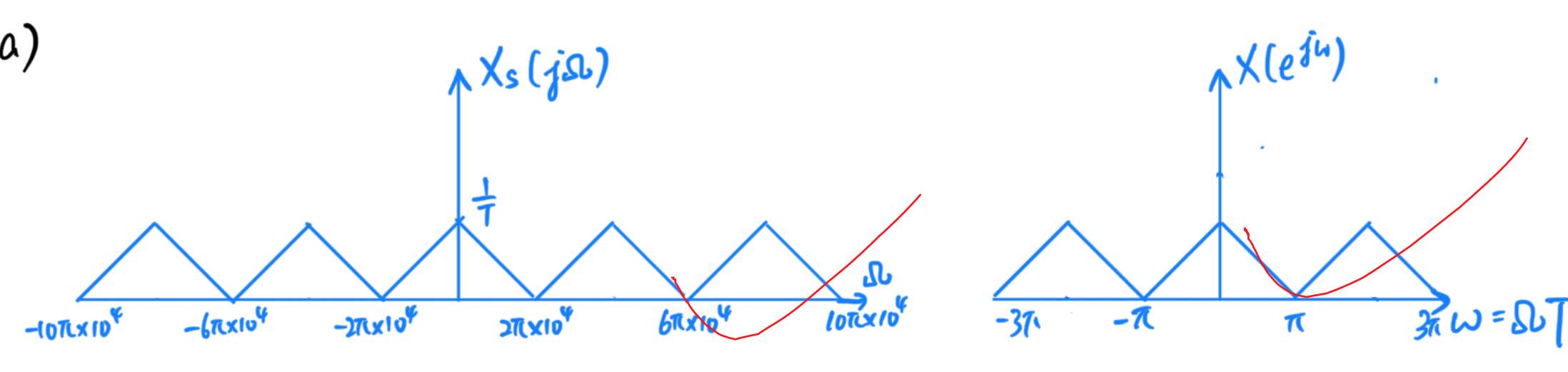
$= \frac{64}{2\pi} \int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} G(e^{j(\omega - \theta)}) d\theta$

$= \frac{64}{2\pi} \int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} G(e^{j(\theta - \omega)}) d\theta$

$= \frac{64}{2\pi} \begin{cases} 0, & |\omega| > \frac{\pi}{4} \\ \frac{\pi}{4} - |\omega|, & |\omega| \leq \frac{\pi}{4} \end{cases}$

$\omega_N = \frac{\pi}{4} < \frac{\pi}{3} \Rightarrow x_r[n] = x[n]$

4.31 (a)



(b) $Y_c(j\Omega) = H_r(j\Omega) Y_c(j\Omega) = H_r(j\Omega) Y(e^{j\Omega T}) = H_r(j\Omega) H(e^{j\Omega T}) X(e^{j\Omega T})$

$= H_r(j\Omega) H(e^{j\Omega T}) X_s(j\Omega) = H_r(j\Omega) H(e^{j\Omega T}) \cdot \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - \frac{2\pi k}{T}))$

$= \begin{cases} 0, & |\Omega| > \frac{\pi}{4T} \\ \sum_{k=-\infty}^{\infty} X_c(j(\Omega - \frac{2\pi k}{T})), & |\Omega| \leq \frac{\pi}{4T} \end{cases}$ ① 混叠可发生在超出 $\frac{\pi}{4} \Delta t$

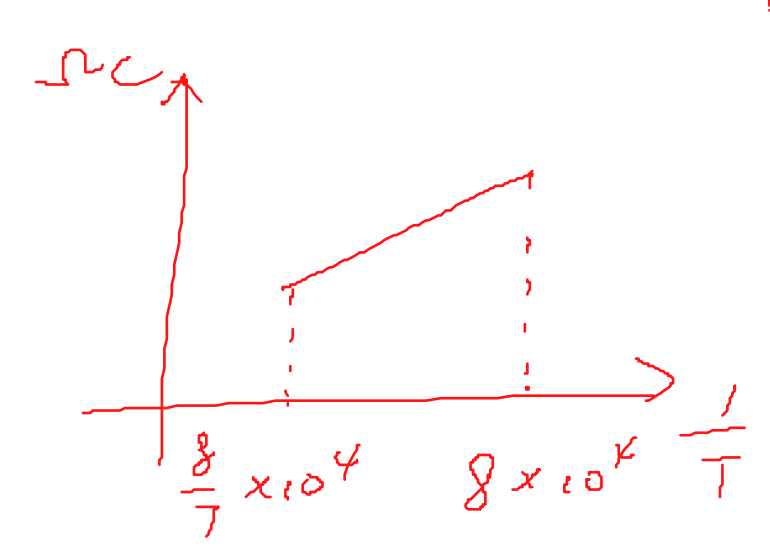
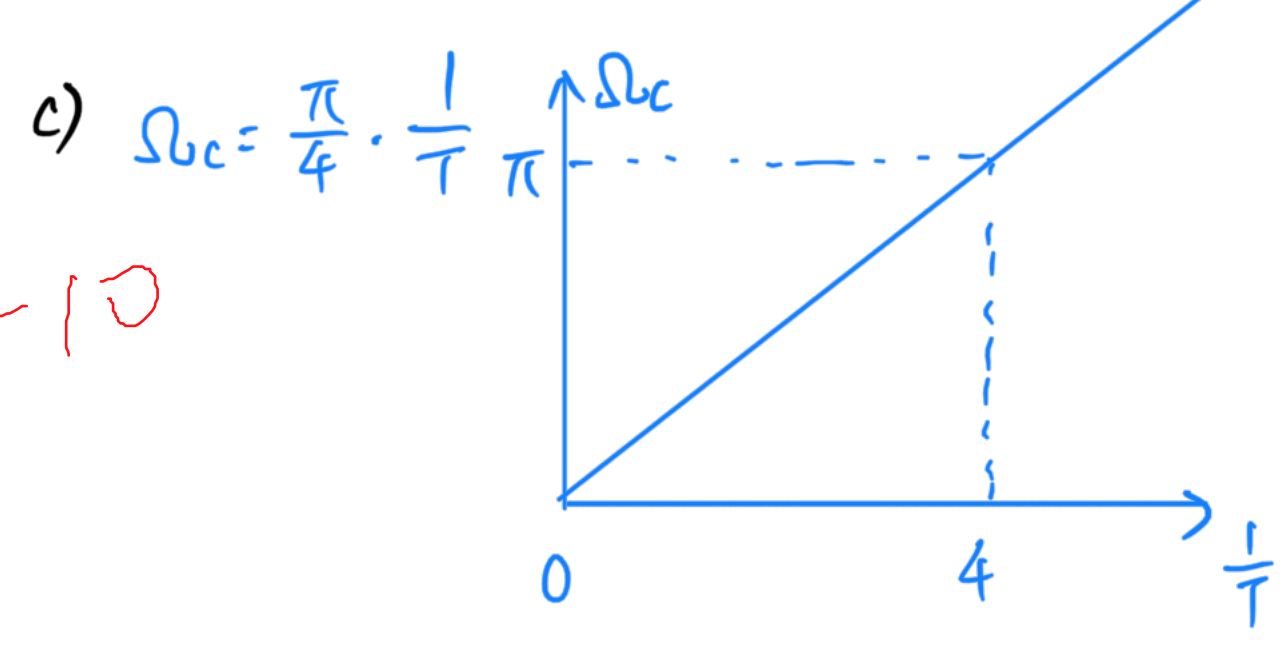
T满足 $\frac{2\pi}{T} \geq 2 \times 2\pi \times 10^4 \Leftrightarrow T \leq \frac{1}{2 \times 10^4} s = 5 \times 10^{-5} s = T_c$

$\frac{2\pi}{T} - 2\pi \times 10^4 \geq \frac{\pi}{4T} \Rightarrow T \leq \frac{2}{8} \times 10^{-4} s$

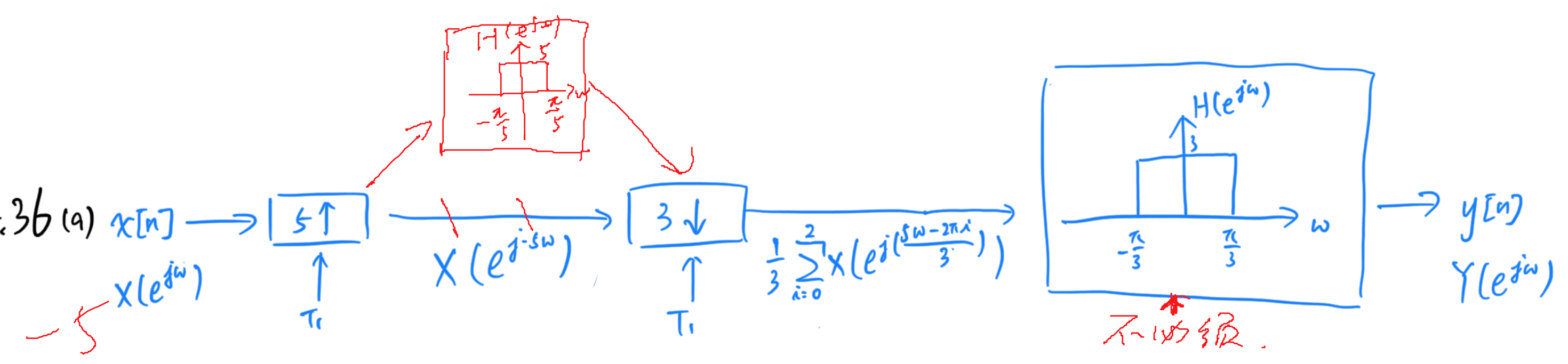
在 a) 中, $\frac{1}{T} = 20kH_z \Leftrightarrow T = 5 \times 10^{-5} s = T_c \Rightarrow$ a) 中结论成立

② 下采样不混叠

$\frac{1}{T} = \frac{4}{\pi} \Omega_c \leq \frac{4}{\pi} \times 2\pi \times 10^4 = 8 \times 10^4$



4.36 (a)



(b) $x[n] = \delta[n] \Leftrightarrow X(e^{j\omega}) = 1 \Leftrightarrow X(e^{j\omega}) = 1 \Rightarrow \frac{1}{3} \sum_{k=0}^2 X(e^{j(\omega - \frac{2\pi k}{3})}) = 1$

$\Rightarrow Y(e^{j\omega}) = \begin{cases} 3, & |\omega| \leq \frac{\pi}{3} \\ 0, & \text{otherwise} \end{cases}$

$\Leftrightarrow y[n] = \frac{1}{2\pi} \int_{-\pi/3}^{\pi/3} 3 \cdot e^{j\omega n} d\omega$

$= \frac{3}{2\pi} \cdot \frac{1}{jn} \cdot j \sin(\pi n/3)$

$= \frac{\sin(\pi n/3)}{\pi n/3} = S_a(n/3)$